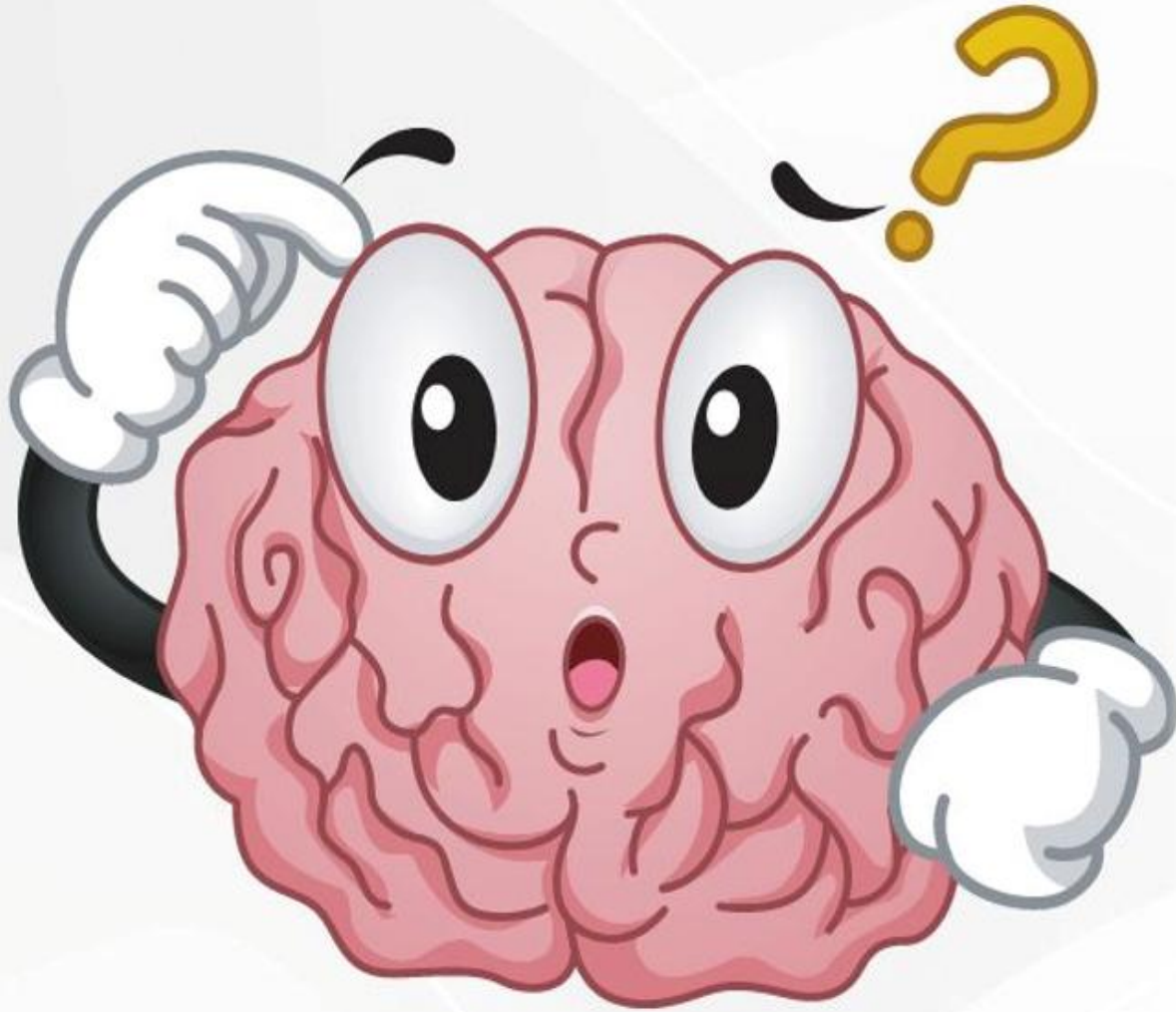




Critical Thinking, Reasoning And Logic

Lesson 3: Types of Arguments:
Deduction and Induction
(Reasoning)

Dr. Mayada Mahmoud

Stanford collage certified TOT

American Board TOT

Dale Carnegie certified Soft Skills

Lesson Objectives:



- **After the successful accomplishment of this lesson, you will be able to:**
 - Understand the meaning, nature, and forms of a deductive argument/ reasoning.
 - Understand the meaning, nature, and forms of an inductive argument/ reasoning.
 - Distinguish deductive arguments from inductive arguments, and vice versa.



“Inductive” **vs.** “Deductive”

- *Inductive* and *deductive* are used in logic, reasoning, and science.
- Scientists use both *inductive* and *deductive* reasoning as part of the [scientific method](#).
- Fictional detectives like Sherlock Holmes are famously associated with methods of *deduction* .



What's the difference between *inductive* and *deductive* ?

- The difference involves whether the reasoning moves from:

1. The general to the specific or

THE
DEDUCTIVE
MIND



2. From the specific to the general.



THE
INDUCTIVE
MIND

What's the difference between *inductive* and *deductive*



Feature	Inductive	Deductive
Direction	Specific → general	General → specific
Conclusion type	Probable (Likely but not certain).	Certain (Logically guaranteed).
Evaluation term	Strong / cogent	Valid / sound
Example	Premise: Every swan I've seen is white. Conclusion: Therefore, all swans are probably white.	Premise1: All humans are mortal. Premise2: Socrates is a human. Conclusion: Therefore, Socrates is mortal.
Definition	It is one where the conclusion is <u>probable</u>, based on the evidence given. Even if the premises <u>are true</u>, the conclusion is <u>not guaranteed</u> to be true.	It is one where the conclusion <u>logically follows</u> from the premises. If the premises <u>are true</u>, the <u>conclusion must be true</u>.

What does *inductive* mean?

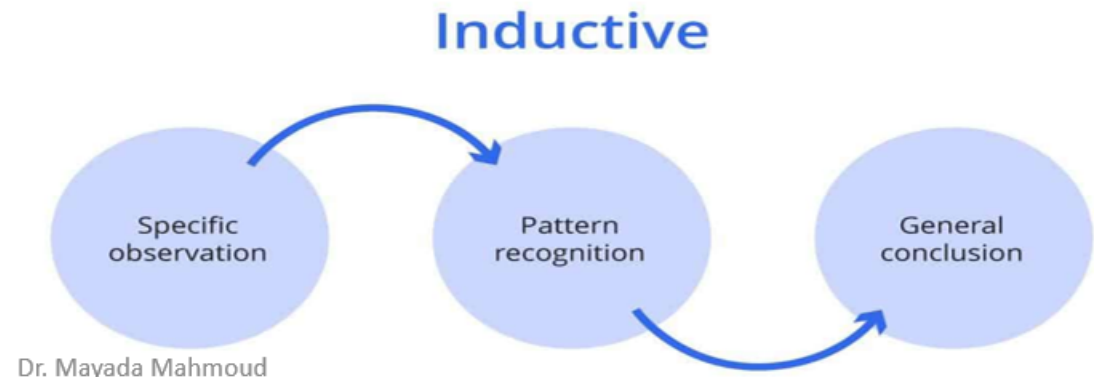
- **Inductive:**

1. Is used to describe reasoning that involves using specific observations, such as **observed patterns**, to make a general conclusion.

- This method is sometimes called induction.

Induction starts with a set of premises, based on experience or evidence.

2. Is a method of drawing conclusions by going from the specific to the general.



What does *inductive* mean?

- In inductive research:

If the argument is strong: it means that the truth of the premise would indicate that the conclusion is correct.

If the argument is weak it means that the logic connecting the premise and the conclusion is unlikely or incorrect.

Exam Treat



- **Inductive Argument — Scientific Reasoning Example**
- **Premise 1:** In numerous controlled experiments, increased carbon dioxide levels have been associated with a rise in global temperatures.
- **Premise 2:** Industrial activity significantly increases atmospheric CO₂ levels.
- **Conclusion:** Therefore, industrial activity probably contributes to global warming.
- **This is inductive:** It generalizes from observed correlations and infers a likely causal relationship.
- It's based on evidence, but it's not 100% certain it's strong, not conclusive.

What does *inductive* mean?



- For example

You go to a cafe every day for a month, and every day, the same person comes at exactly 11 am and orders a cappuccino.

The observation :is that this person has come to the cafe at the same time and ordered the same thing every day during the period observed.

The conclusion : this person always comes to the cafe at the same time and orders the same thing.



Exam Treat



1. *The known fish species in this genus have yellow fins.*

- **Premise/ observation :** All known fish species in this genus have yellow fins.
- **Conclusion:** Any newly discovered species in the genus is likely to have yellow fins.

2. *This volcano has erupted about every 500 years for the last 1 million years. It last erupted 499 years ago.*

- **Premises/observation:** This volcano has erupted about every 500 years for the last 1 million years. It last erupted 499 years ago.
- **Conclusion:** It will erupt again soon.

Exam Treat



- **Inductive Argument — Forensic Reasoning Example**
- **Premise 1:** The suspect's fingerprints were found on the murder weapon.
- **Premise 2:** The suspect had no alibi for the time of the crime.
- **Premise 3:** The suspect had a motive to harm the victim.
- **Conclusion:** Therefore, the suspect is probably guilty of the crime.
- **This is inductive** because it builds a probable conclusion based on circumstantial evidence.
- A jury might convict based on this, but it's still not deductively certain.

Exam Treat



- **Inductive Argument — Economic Forecasting**
- **Premise 1:** Every time interest rates have risen quickly in the past decade, a recession followed within 12 months.
- **Premise 2:** Interest rates have just risen sharply again.
- **Conclusion:** Therefore, the economy will probably enter a recession within the next year.
- Based on past patterns, so it's inductive.
- Forecasts are educated guesses, not guarantees.

Exam Treat



1. *All plants with rainbow berries are poisonous. This plant has rainbow berries.*

- **Premises:** All plants with rainbow berries are poisonous. This plant has rainbow berries.
- **Conclusion:** This plant is poisonous.

2. *I am lactose intolerant. Lactose intolerant people get sick when they consume dairy. This milkshake contains dairy.*

- **Premises:** I am lactose intolerant. Lactose intolerant people get sick when they consume dairy. This milkshake contains dairy.
- **Conclusion:** I will get sick if I drink this milkshake.

Exam Treat



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Exam Treat



- **Deductive Argument — Mathematics in Logic Puzzles**

- **Premise 1:** In this puzzle, only one person can be the impostor.
- **Premise 2:** If Alex is innocent, then Taylor must be lying.
- **Premise 3:** Taylor is not lying.
- **Conclusion:** Therefore, Alex is not innocent.
- **This uses deductive logic** through a process of elimination and conditional statements.

Exam Treat



- **Deductive Argument — Philosophical Ethics Example**
- **Premise 1:** If an action is morally wrong, then it should not be legally permissible.
- **Premise 2:** Torture is morally wrong.
- **Conclusion:** Therefore, torture should not be legally permissible.
- This argument is logically valid — if you accept the moral principle and the judgment about torture, the conclusion follows.
- Whether it's sound depends on whether you agree with the premises.

Inductive **vs.** Deductive Reasoning

Inductive	Deductive
Is often called a “ bottom-up approach” because you start with an observation, detect patterns, formulate a hypothesis, and reach a conclusion/theory.	Is often called a “ top-down approach” because you start with a theory, narrow it down to a hypothesis, observe the hypothesis, and ultimately reach a logical affirmation.

Let's Practice Exam



- Determine whether the following argument is inductive or deductive:
 1. *"The last manager was honest. The current manger is honest. All manager are honest."*
 2. *"Every word has the letter e in it. Therefore, your name has the letter e in it."*

Answer: 1- inductive.

2- deductive .

